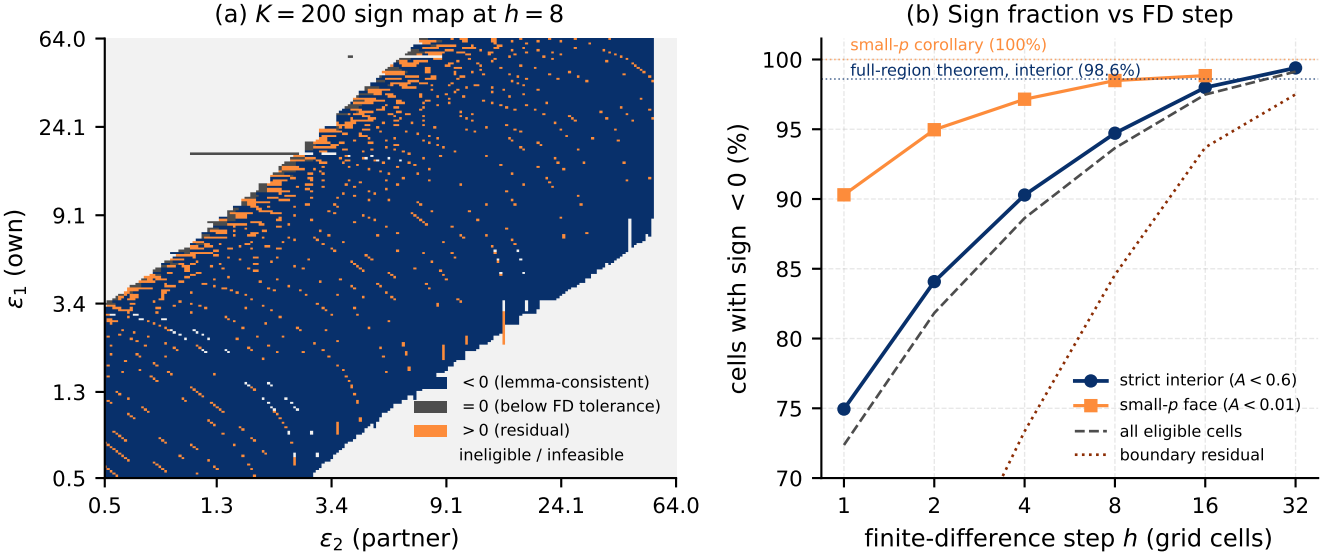
(a) $K = 200$ sign map at $h = 8$
64.0
24.1
9.1
3.4
1.3
0.5
$\varepsilon_1$ (own)
0.5
1.3
3.4
9.1
24.1
64.0
$\varepsilon_2$ (partner)
< 0 (lemma-consistent)
= 0 (below FD tolerance)
> 0 (residual)
ineligible / infeasible
(b) Sign fraction vs FD step
small-$p$ corollary (100%)
full-region theorem, interior (98.6%)
100
95
90
85
80
75
70
cells with sign < 0 (%)
strict interior ($A < 0.6$)
small-$p$ face ($A < 0.01$)
all eligible cells
boundary residual
1
2
4
8
16
32
finite-difference step $h$ (grid cells)